Code No: 138FY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
ELECTRONIC MEASURING INSTRUMENTS
(Common to CE, EEE, ME, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) What is the type of response in electromechanical recorders? Derive the necessary Equations.
b) A 153± 25% resistance is connected to a power supply source operating at 313V dc. What range of current would flow if the resistor varied over the range of ±13% of its expected value? What is the range of error in the current? [9+6]

2.a) Explain the working of Sine wave generator that works in the frequency range of 20Hz-20KHz.
b) With a neat block diagram, explain the working of generators that are used in transient response testing of ampifiers. [8+7]

3.a) With a neat diagram, explain how DC as well as AC current and voltage can be measured.
b) The Lissajous pattern on a CRO is stationary and has five horizontal and three vertical tangencies. The frequency of the horizontal input is 1000Hz. Determine the frequency of vertical input. [8+7]

4.a) With a neat diagram, explain about the recorders that are well suited for low frequency AC Inputs obtained from quantities varying slowly at frequencies upto 100c/s, or in special cases upto 1000c/s.
b) Explain how high frequency signals can be recorded as well as the data can be replayed almost indefinite number of times. [6+9]

5.a) With a neat diagram, explain about Piezoelectric transducer and also derive the expression For coupling coefficient.
b) Explain how displacement can be measured using a neat diagram. [8+7]

6.a) What is the significance of repeatability and reproducibility?
b) With a neat block diagram, explain the operation of generator which can generate any arbitrary wave. [7+8]

JNTUH Used Papers

7.a) A series type ohmmeter is designed to operate with a 6V battery. The meter movement has an internal resistance of 2KΩ and requires a current of 120 μA for full scale deflection. The value of R_1 is 49K.

i) Assuming the battery voltage has fallen to 5.9V, calculate the value of R_2 required to "0" the meter.

ii) Under the condition mentioned in (i), an unknown resistance is connected to the meter, causing a 60% deflection. Calculate the value of the unknown resistance.

b) A 500μA meter movement with an internal resistance of 800Ω is to be used in 0-200mA current meter. Find the value of the required shunt. [9+6]

8.a) Name and explain various static characteristics of measuring systems.

b) Write a short note on AF signal generator. [8+7]

--ooOoo--

JNTUH Used Papers